Laboratory 08

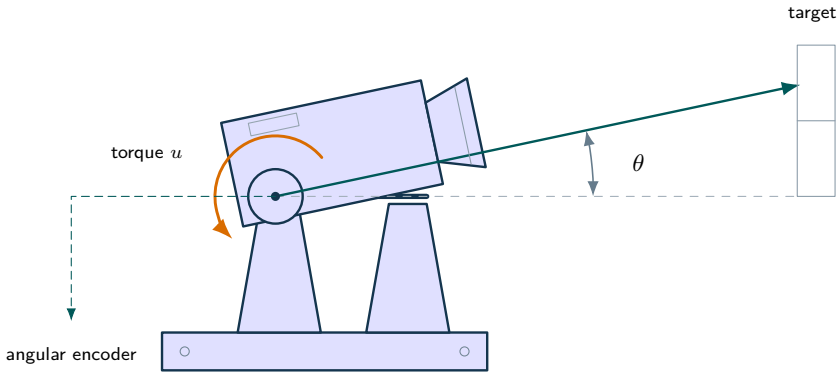
Lead-lag design for a normalized camera axis

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal camera-axis plant: $G(s) = 1/[s(s+1)]$. Design crossover is 3 rad/s and desired phase margin is 55°. Use unrounded compensator parameters in subsequent calculations.

1. A camera axis with simultaneous speed and phase requirements

The physical sketch shows a single rotating camera axis with a position sensor. The normalized transfer is stipulated; inertia, drive bandwidth and unit scaling are already aggregated into it.



Gimbal axis with angular readout

Normalized model $G(s) = 1/[s(s+1)]$; geometry is illustrative

Coordinate and actuation
Positive angle is measured from the dashed reference ray. Torque acts about the marked gimbal pivot.

Normalized axis
$G(s) = \dfrac{1}{s(s+1)}$.
Lead supplies phase; lag changes low-frequency gain and the ramp error.

Limits of the view
Encoder and lens establish the signal meaning. Camera inertia, gearing and dimensions do not identify this stipulated plant.

A camera gimbal illustrates angular tracking and disturbance rejection. Its normalized plant is stipulated, with no claim that geometry alone determines its two poles.

(a) The normalized plant is $G(s) = 1/[s(s+1)]$. At unity controller gain, derive the closed-loop polynomial, gain crossover and phase margin.

(b) A redesign targets crossover 3 rad/s and phase margin 55 degrees. Compute the plant phase at that frequency and the phase the lead section must provide. Explain why changing scalar gain alone cannot achieve both targets.

(c) Suppose an omitted actuator lag $1/(0.1s+1)$ is added to the plant. At the fixed 3 rad/s crossover, find the scalar gain needed for unity magnitude, the resulting phase margin, and the lead phase now required for the 55° target. Compare with the original plant.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. Realizing and normalizing a lead network

The desired phase improvement must be produced by a realizable pole–zero pair and then normalized at the selected frequency.

(a) For $C_{lead} = K(Ts + 1)/(\alpha Ts + 1)$, center maximum phase lead at 3 rad/s and supply the phase found in Problem 1(b). Derive α and T and locate the controller zero and pole.

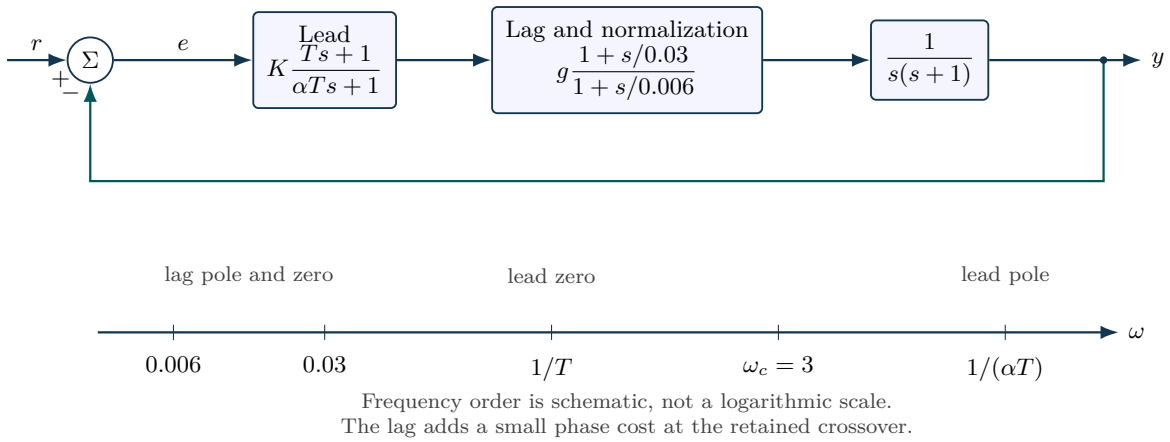
(b) Find K so the lead-compensated magnitude equals one at 3 rad/s. Show explicitly why the magnitude of the lead section cannot be ignored.

(c) An implementation accidentally exchanges the calculated lead pole and zero. Determine the sign of its maximum phase contribution and explain why retuning scalar gain cannot repair that phase error.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. Improving ramp tracking without silently relaxing phase specifications

Add a lag network to improve low-frequency tracking while preserving crossover. Carry the gain normalization through all steady-error calculations.



Frequency order is schematic, not a logarithmic scale.
The lag adds a small phase cost at the retained crossover.

Lead and lag sections are drawn separately so their gain normalization and frequency ordering can be analyzed independently.

(a) For the stable lead-compensated loop, calculate its velocity constant and unit-ramp error. State why the DC gain of G itself cannot be inserted as a finite number.

(b) Add $F(s) = (1 + s/0.03)/(1 + s/0.006)$ and multiply the controller by g so crossover remains at 3. Derive g, the new velocity constant and the new unit-ramp error.

(c) Require unit-ramp error no greater than 0.05 and phase margin at least 55 degrees. Using the resulting design, check both requirements together rather than rounding the margin.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. Spending the remaining phase reserve on a real communication link

Evaluate the final lead–lag design after a pure communication delay is introduced. Keep strict stability and a desired robustness margin as separate requirements.

(a) Compute the lag section phase at the retained crossover and the resulting phase margin. Explain why a lag network can improve low-frequency tracking while slightly reducing margin.

(b) Assume the final lead-lag loop has the single crossover 3 rad/s and phase margin from Problem 4(a). Calculate its first constant-delay boundary and the delay allowance if at least 40 degrees of margin must remain. State the assumptions.

(c) A proposed link adds 0.09 s of constant pure delay. Does the model retain 40 degrees of margin? Is failure of that requirement equivalent to instability under the stated single-crossover assumptions?

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. A stable compensated output can still demand noisy actuator effort

Check algebraic stability and then trace measurement noise to both physical output and controller output. These are different performance channels.

(a) Derive the lead-only characteristic polynomial and use the cubic Routh inequality to check stability with the design formulas. Do not infer stability solely from an intended phase target.

(b) Let measurement be $y+n$ and feedback error be $r-y-n$. Derive the noise-to-output and noise-to-control transfers. Explain a high-frequency consequence of lead compensation.

(c) For the final controller $C = gK(Ts+1)(1+s/0.03)/[(\alpha Ts+1)(1+s/0.006)]$, find its high-frequency gain and the corresponding limiting noise-to-control and noise-to-output gains.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.